

Contoso Systems Technical Report

Prepared by Engineering Analysis Group

For Contoso Systems

Project: Edge Monitoring Deployment

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1. Executive Summary

This report presents the findings of a comprehensive performance audit conducted on the organisation's core data-processing infrastructure. Key findings include a 38% over-provisioning of compute resources, significant I/O bottlenecks in the primary database tier, and an opportunity to reduce annual infrastructure costs by approximately \$240 000 through targeted optimisations.

2. Background

The data-processing platform was originally deployed in 2019 to handle an anticipated throughput of 10 000 transactions per second. Since then, organic growth and three major product launches have increased peak load to 27 000 TPS, exceeding the original design envelope by nearly a factor of three. This chapter describes the current architecture, its evolution, and the symptoms that prompted the present audit.

3. Analysis

Provide detailed analysis, calculations, and assessments. Table 3.1 summarises the key performance metrics observed during the two-week monitoring period.

Table 3.1: Infrastructure performance summary..

Metric	Target	Observed	Status
Avg. response time (ms)	< 200	312	Over
P99 latency (ms)	< 500	1 240	Over
CPU utilisation (%)	60–80	43	Under
Memory utilisation (%)	70–85	62	Under
Disk I/O wait (%)	< 10	28	Over

The elevated disk I/O wait is the primary driver of latency degradation. The storage subsystem saturates at approximately 18 000 IOPS, whereas peak demand reaches 24 000 IOPS. Figure 3.1 illustrates the IOPS demand profile over a representative 24-hour period.

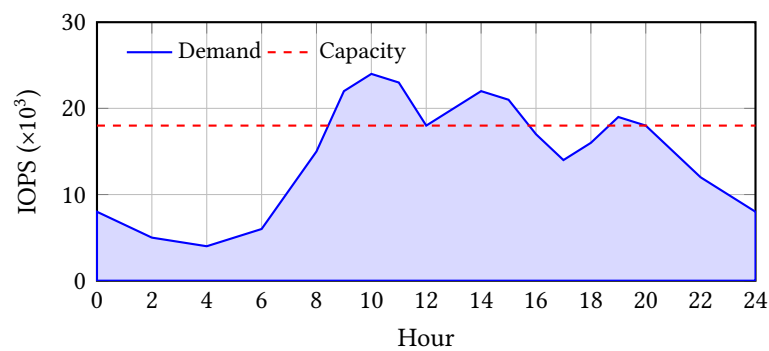


Figure 3.1: Disk IOPS demand versus capacity over a typical day..

4. Recommendations

Based on the analysis above, we recommend the following actions, prioritised by impact and implementation effort:

1. **Migrate to NVMe-backed storage** — Expected to raise the IOPS ceiling to 80 000, eliminating the current bottleneck.
2. **Right-size compute instances** — Reducing CPU and memory allocations by 30–35% on 40% of workloads without impacting SLAs.
3. **Implement connection pooling** — Reuse database connections to reduce per-query overhead by an estimated 15%.
4. **Introduce read replicas** — Offload reporting queries to dedicated read replicas to isolate analytical workloads.

A. Supporting Data

Include raw data, extended monitoring logs, and cost projections supporting the recommendations in this report.